Extreme Estimation

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1 Introduction and examples

Motivation for extreme estimation framework: lack of "closed form"

- Both the OLS and IV estimators have "closed form", e.g. $\hat{\beta}_{ols} = (X'X)^{-1}(X'Y)$. That is, given data, the estimates can be computed using a known formula.
- Closed form makes the asymptotic properties of an estimator easy to derive.
- For nonlinear models, estimators usually do not have closed form.
- Even if the underlying model is linear, estimators may do not have closed form.

Median regression example

- Consider median regression,

$$\text{Med}(y|x) = x'\theta = \theta_1 + \theta_2 x_2 + \cdots + \theta_K x_K.$$

  Here $\text{Med}(y|x)$ is the conditional *median* of y given x.

- Without additional assumptions, OLS does not consistently estimate the θ_j. Consider a simple case $\text{Med}(y|x) = \theta_1$. The OLS converges to $\text{E}(y)$, which does not have to be its median.

- But we will show *least absolute deviations* (LAD) does. Let $\hat{Q}_n(b) = n^{-1}\sum_{i=1}^n |y_i - x_i'b|$. The LAD estimator solves the minimization problem,

$$\min_{b \in \mathbb{R}^{K+1}} \hat{Q}_n(b). \tag{1}$$

  This problem does not have closed form solution.

- The difficulty in deriving the asymptotic properties of the LAD estimator is the lack of closed form.

- Let $\hat{\theta}_{LAD}$ be a solution of eq. (1). In principle, we can write

$$\hat{\theta}_{LAD} = g_n(z_1, \ldots, z_n), \quad z_i = (x_i', y_i)',$$

  for some function g_n. The problem is we do not know g_n.

- Question: If we do not know g_n, how do we study the large-sample (asymptotic) properties of $\hat{\theta}_{LAD}$?

- Extreme estimation theory provides a framework of studying the asymptotic property of an estimator based on the objective function.

- It starts with the observation that the sample objective function $\hat{Q}_n(b)$ usually converges to some population objective function $Q_0(b)$ that is minimized at the true value.

- For each b, $\hat{Q}_n(b) \equiv n^{-1}\sum_{i=1}^n |y_i - x_i'b|$ is an average of iid random function, $q(z_i, b) \equiv |y_i - x_i'b|$, for which we can apply the LLN.

- It would not be a surprise that $\hat{Q}_n(b) \to_p Q_0(b) \equiv \text{E}(q(z_i, b))$ for all b.

- The true θ_0 also minimizes $Q_0(b)$.

- If $\hat{Q}_n(\theta) \to_p Q_0(\theta)$ for all θ, it is not crazy to expect that $\hat{\theta}_n \to_p \theta_0$.

Nonlinear least square example

- Suppose for $y \geq 0$, and we specify an exponential conditional mean model:

$$\text{E}(y \mid x) = \exp(x'\theta).$$

- OLS using $\log(y)$ as the dependent variable does not work, if $\Pr(y = 0) > 0$.
- Letting $\hat{Q}_n(b) = n^{-1} \sum_{i=1}^n (y_i - \exp(x_i' b))^2$, *nonlinear least squares* (NLS) estimator $\hat{\theta}_{nls}$ solves
$$\min_b \hat{Q}_n(b).$$
- NLS $\hat{\theta}_{nls}$ does not have a closed form. But it minimizes a function that is an average of i.i.d. random functions of b too.
- We expect $\hat{Q}_n(\theta)$ converges to $Q_0(\theta) = \mathrm{E}([y_i - \exp(x_i'\theta)]^2)$.
- We have the following useful result: assume $\mathrm{E}(y^2) < \infty$ and let $\mathrm{E}(y \,|\, x) = \mu_0(x)$. Then among all functions $\mu(x)$ with $\mathrm{E}\big(\mu(x)^2\big) < \infty$,
$$\mathrm{E}([y - \mu_0(x)]^2) \leq \mathrm{E}([y - \mu(x)]^2)$$
It follows that if $\mathrm{E}(y \mid x) = m(x, \theta_0)$, then
$$\mathrm{E}([y - m(x, \theta_0)]^2) \leq \mathrm{E}([y - m(x, \theta)]^2), \text{ all } \theta \in \Theta.$$
- That is, $Q_0(\theta) = \mathrm{E}([y - m(x, \theta_0)]^2)$ is minimized at θ_0.
- So we hope $\hat{\theta}_{nls}$ converges to θ_0.

GMM example

- Suppose that there is a "moment function" vector $g(z, \theta)$ such that $\mathrm{E}(g(z, \theta_0)) = 0$ for some constants θ_0.
- A GMM estimator aims to estimate θ_0 by minimizing
$$\min_\theta \hat{Q}_n(\theta) \equiv= \left(n^{-1} \sum_{i=1}^n g(z_i, \theta) \right)' \hat{W} \left(n^{-1} \sum_{i=1}^n g(z_i, \theta) \right),$$
where $\hat{W}$ is a weighting matrix.
- Excepting for special g, we do not have a closed form solution of the above problem.
- It would not be a surprise that
$$\hat{Q}_n(\theta) \to_p Q_0(\theta) \equiv (\mathrm{E}(g(z_i, \theta)))' W_0 (\mathrm{E}(g(z_i, \theta))).$$

2 Consistency of extreme estimators

Notation

- Let θ denote a finitely dimensional generic vector of parameters, and let θ_0 denote the true value.
- Extreme estimator $\hat{\theta}_n$ of θ_0 is from

$$\hat{\theta}_n = \arg\min_{\theta \in \Theta} \hat{Q}_n(\theta).$$

  Θ is the parameter space, which contains all possible candidates of θ_0.
- If $\hat{Q}_n(\theta)$ has a sample average form, that is if $\hat{Q}_n(\theta) = n^{-1} \sum_{i=1}^n q(z_i, \theta)$ for some q, the estimator is typically called M estimator.
- In the next few lectures, we use M estimation theory frequently.

Consistency theorem

Theorem 1. *Let* $\left\{\hat{Q}_n(\theta)\right\}$ *be a sequence of random functions on a compact set* $\theta \subset \mathbb{R}^p$ *such that for a continuous real function* $Q_0(\theta)$ *on* θ*,*

$$\hat{Q}_n(\theta) \to Q_0(\theta) \quad \textit{a.s. (in pr.)} \ \textbf{\textit{uniformly}} \ \textit{on } \Theta. \qquad \text{(convergence of criterion function)}$$

Let $\hat{\theta}_n$ *be any random vector in* Θ *satisfying*

$$\hat{Q}_n(\hat{\theta}_n) = \inf_{\theta \in \Theta} \hat{Q}_n(\theta) \qquad \text{(extreme in sample criterion)}$$

and let θ_0 *be a* ***unique*** *point in* Θ *such that*

$$Q_0(\theta_0) = \inf_{\theta \in \Theta} Q_0(\theta). \qquad \text{(extreme in population criterion)}$$

Then $\hat{\theta}_n \to \theta_0$ *a.s. (in pr.)*

Uniform convergence in probability: $\hat{Q}_n(\theta)$ converges uniformly in probability to $Q_0(\theta)$ means

$$\sup_{\theta \in \Theta} \left| \hat{Q}_n(\theta) - Q_0(\theta) \right| \to_p 0.$$

Sketch of proof

- First, we show $Q_0(\hat{\theta}_n) - Q_0(\theta_0) \to_p 0$.

$$\begin{aligned} 0 &\leq Q_0(\hat{\theta}_n) - Q_0(\theta_0) \\ &= Q_0(\hat{\theta}_n) - \hat{Q}_n(\hat{\theta}_n) + \hat{Q}_n(\hat{\theta}_n) - Q_0(\theta_0) \\ &\leq Q_0(\hat{\theta}_n) - \hat{Q}_n(\hat{\theta}_n) + \hat{Q}_n(\theta_0) - Q_0(\theta_0) \\ &\leq 2 \cdot \sup_{\theta \in \Theta} \left| \hat{Q}_n(\theta) - Q_0(\theta) \right| = o_p(1). \end{aligned}$$

- By continuity of $Q_0(\cdot)$, $Q_0(\hat{\theta}_n) - Q_0(\theta_0) \to_p 0$ implies $\hat{\theta}_n \to_p \theta_0$.

Uniform convergence

- When $\hat{Q}_n(\theta)$ is sample average, the *uniform law of large numbers* (ULLN) is a powerful device for establishing the uniform convergence.

- Here is the ULLN from Lemma 2.4 of Newey-McFadden.

  Lemma 1. *If the data are iid, Θ is compact, $a(z_i, \theta)$ is continuous at each $\theta \in \Theta$ with probability one, and there is $d(z)$ with $\|a(z, \theta)\| \leq d(z)$ for all $\theta \in \Theta$ and $\mathrm{E}(d(z)) < \infty$, then $\mathrm{E}(a(z, \theta))$ is continuous and $\sup_{\theta \in \Theta} \|n^{-1} \sum_{i=1}^n a(z_i, \theta) - \mathrm{E}(a(z, \theta))\| \to_p 0$.*

- ULLN is enough for many examples of extreme estimator, including MLE, NLS, and GMM.

- A more general theorem is in Newey (1991, Uniform Convergence in Probability and Stochastic Equicontinuity, *Econometrica*), which shows that compactness, pointwise convergence and stochastic equicontinuity implies uniform convergence.

"Uniquely minimized at θ_0" is an identification condition

Example 1. *Suppose $x > 0$ is a scalar, $y \geq 0$, and $m(x, \theta) = \theta_1 + \theta_2 x^{\theta_3}$, where the parameter spaces is $\Theta = [0, \infty) \times [0, \infty) \times \mathbb{R}$. If $\theta_{02} = 0$, $\mathrm{E}(y \mid x) = \mathrm{E}(y) = \theta_{01}$, then any θ of the form $(\theta_{01}, 0, \theta_3)'$ minimizes the mean squared error. What would NLS estimate for θ_{03}?*

- We cannot identify θ_{03}.

Identification assumption for NLS

- We need an identification assumption for NLS that rules out the previous situation.

  Assumption 1. *Let* $m(x,\theta) = \mathrm{E}(y|x)$*, and* $\mathrm{E}([m(x,\theta_0)-m(x,\theta)]^2) > 0$ *for all* $\theta \in \Theta$*,* $\theta \neq \theta_0$*.*

- Assumption 1 plays the role of a rank condition. In the linear case, $m(x,\theta) = x'\theta$, and then

$$m(x,\theta_0) - m(x,\theta) = (\theta_0 - \theta)' x x' (\theta_0 - \theta).$$

$$\mathrm{E}\big((m(x,\theta_0) - m(x,\theta))^2\big) = (\theta_0 - \theta)' \,\mathrm{E}(xx')(\theta_0 - \theta).$$

  For the last expression to be positive for all $\theta \neq \theta_0$, we need $\mathrm{E}(xx')$ to have full rank.

Convergence of estimated sample averages

- Lemma 4.3 of Newey and McFadden is a very useful result for estimating asymptotic variance.

  Lemma 2. *If* z_i *is iid,* $a(z,\theta)$ *is continuous at* θ_0 *with probability one, and there is a neighborhood* $\mathcal{N}$ *of* θ_0 *such that* $\mathrm{E}(\sup_{\theta\in\mathcal{N}}\|a(z,\theta)\|) < \infty$*, then for any* $\hat{\theta}_n \to_p \theta_0$*,*

$$n^{-1}\sum_{i=1}^{n} a(z_i,\hat{\theta}_n) \to_p \mathrm{E}(a(z,\theta_0)).$$

- This lemma can be used to show the heteroskedasticity robust variance estimator in linear regression.

Proof of lemma 2. By consistency of $\hat{\theta}_n$, there is $\delta_n \to 0$ such that $\|\hat{\theta}_n - \theta_0\| \leq \delta_n$ w.p.a.1. Let $\Delta_n(z) = \sup_{\|\theta-\theta_0\|\leq\delta_n}\|a(z,\theta)-a(z,\theta_0)\|$. By continuity of $a(z,\theta)$ at θ_0, $\Delta_n(z) \to 0$ with probability one. By the dominance condition, $\mathrm{E}(\sup_{\theta\in\mathcal{N}}\|a(z,\theta)\|) < \infty$, we first have that for n large enough (so that the set $\{\theta : \|\theta-\theta_0\| \leq \delta_n\} \subset \mathcal{N}$), $\Delta_n(z) \leq 2\sup_{\theta\in\mathcal{N}}\|a(z,\theta)\|$. Then by the dominated convergence theorem, $\mathrm{E}(\Delta_n(z)) \to 0$.

By the Markov inequality, we have

$$\Pr\left(\left|n^{-1}\sum_{i=1}^{n}\Delta_n(z_i)\right| > \varepsilon\right) \leq \mathrm{E}(\Delta_n(z))/\varepsilon \to 0.$$

We conclude $n^{-1}\sum_{i=1}^{n}\Delta_n(z_i) \to_p 0$.

We have

$$\left\| n^{-1}\sum_{i=1}^{n} a(z_i,\hat{\theta}_n) - n^{-1}\sum_{i=1}^{n} a(z_i,\theta_0) \right\| \leq n^{-1}\sum_{i=1}^{n} \|a(z_i,\hat{\theta}_n) - a(z_i,\theta_0)\|$$

W.p.a.1,

$$n^{-1}\sum_{i=1}^{n}\|a(z_i,\hat{\theta}_n)-a(z_i,\theta_0)\| \leq n^{-1}\sum_{i=1}^{n}\sup_{\|\theta-\theta_0\|\leq\delta_n}\|a(z,\theta)-a(z,\theta_0)\| = n^{-1}\sum_{i=1}^{n}\Delta_n(z_i) = o_p(1).$$

The first inequality follows from w.p.a.1 $\|\hat{\theta}_n-\theta_0\| \leq \delta_n$. By the LLN, $n^{-1}\sum_{i=1}^{n} a(z_i,\theta_0) \to_p \mathrm{E}(a(z,\theta_0))$. The lemma follows from the triangle inequality.

Dominated convergence theorem. Let $\{f_i\}$ be a sequence of integrable functions on (Ω,Σ,μ) and suppose that

$$\lim_{n\to\infty} f_n(x) = f(x) \quad \text{almost everywhere -}\mu.$$

Further suppose that there is an integrable $G \geq 0$ such that $|f_n(x)| \leq G(x)$ for any n. Then $|f(x)| \leq G(x)$ and

$$\lim_{n\to\infty}\int_\Omega f_n(x)\ \mathrm{d}\,\mu(x) = \int_\Omega f(x)\ \mathrm{d}\,\mu(x).$$

3 Asymptotic normality of extreme estimator

Main result

Theorem 2. *Let the conditions of theorem 1 hold and assume in addition that:*

1. *Θ is convex and θ_0 is an interior point of Θ;*
2. *the first order and second order derivatives of $\hat{Q}_n(\theta)$ are well defined as vector and matrix of random functions on Θ;*
3. *$\sqrt{n}\partial\hat{Q}_n(\theta_0)/\partial\theta \to_d \mathcal{N}(0,B)$, where B is a positive semi-definite matrix;*
4. *for i,j, $\partial^2\hat{Q}_n(\theta)/\partial\theta_i\partial\theta_j \to_p \partial^2 Q_0(\theta)/\partial\theta_i\partial\theta_j$ uniformly on a neighborhood of θ_0, where the limit function involved is continuous in θ_0;*
5. *the matrix $A = \partial^2 Q_0(\theta_0)/\partial\theta\partial\theta'$ is non-singular.*

Then $\sqrt{n}(\hat{\theta}_n - \theta_0) \to_d \mathcal{N}(A^{-1}BA^{-1})$.

Useful matrix concept

- Let θ be p dimensional vector. Let $a(\theta)$ be a real valued function of θ. Let $f^k(\theta) = (f_1(\theta), \ldots, f_k(\theta))'$ be a k dimensional vector of functions of θ.
- *Gradient* is a $p \times 1$ vector:

$$\nabla a = (\partial a/\partial\theta_1, \partial a/\partial\theta_2, \ldots, \partial a/\partial\theta_k)'.$$

- *Jacobian* matrix is a $k \times p$ matrix, that is $\dim(f) \times \dim(\theta)$:

$$\nabla f = \begin{bmatrix} \partial f_1/\partial\theta_1 & \ldots & \partial f_1/\partial\theta_p \\ \vdots & \vdots & \vdots \\ \partial f_k/\partial\theta_1 & \ldots & \partial f_k/\partial\theta_p \end{bmatrix}$$

- *Hessian* matrix is a $p \times p$ matrix, that is $\dim(\theta) \times \dim(\theta)$:

$$\begin{bmatrix} \partial^2 a/\partial^2\theta_1 & \ldots & \partial^2 a/\partial\theta_1\partial\theta_p \\ \vdots & \vdots & \vdots \\ \partial^2 a/\partial\theta_p\partial\theta_1 & \ldots & \partial^2 a/\partial^2\theta_p \end{bmatrix}$$

- Hessian matrix is the Jacobian of gradient.

Proof of asymptotic normality in M estimation case

- We show the proof in M estimation case.
- In the M estimation, we minimize an objective function $\hat{Q}_n(\theta) = n^{-1}\sum_{i=1}^n q(z_i, \theta)$.
- For the minimization problem, we are interested in its FOC. Let $s(z, \theta) = \nabla_\theta q(z, \theta)$ be the gradient of q w.r.t. θ.
- $s(z, \theta)$ is often called **score function** following the convention of MLE.
- Because θ_0 is in the interior of Θ and $\hat{\theta}_n \to_p \theta_0$, we know that $\hat{\theta}_n$ falls in the interior of Θ with probability one.
- Since $\hat{\theta}_n \in \text{int}(\Theta)$, as a minimizer, $\hat{\theta}_n$ satisfies

$$\sum_{i=1}^n s(z_i, \hat{\theta}_n) = 0.$$

  This is a set of $p = \dim(\theta)$ equations.

- Assuming that q is twice continuously differentiable, each component of $s(z,\theta)$ is continuously differentiable.

- We then can apply the mean value theorem,

$$0 = \sum_{i=1}^{n} s(z_i, \hat{\theta}_n) = \sum_{i=1}^{n} s(z_i, \theta_0) + \left(\sum_{i=1}^{n} \nabla_\theta s(z_i, \theta_*) \right) (\hat{\theta}_n - \theta_0).$$

  Here θ_* is on the line segment between $\hat{\theta}_n$ and θ_0, hence $\theta_* \to_p \theta_0$.

- Write the Hessian matrix

$$H(z,\theta) = \nabla_\theta s(z,\theta) = \frac{\partial^2 q(z,\theta)}{\partial\theta\partial\theta'}$$

- Using the Hessian matrix notation, we have

$$0 = \sum_{i=1}^{n} s(z_i, \hat{\theta}_n) = \sum_{i=1}^{n} s(z_i, \theta_0) + \left(\sum_{i=1}^{n} H(z_i, \theta_*) \right) (\hat{\theta}_n - \theta_0)$$

- So

$$0 = n^{-1/2} \sum_{i=1}^{n} s(z_i, \theta_0) + \left(n^{-1} \sum_{i=1}^{n} H(z_i, \theta_*) \right) \sqrt{n}(\hat{\theta}_n - \theta_0)$$

- By $\theta_* \to_p \theta_0$, we conclude

$$n^{-1} \sum_{i=1}^{n} H(z_i, \theta_*) \to_p \mathrm{E}(H(z, \theta_0)) \equiv A,$$

  by lemma 2.

- Assuming A is nonsingular (hence $n^{-1} \sum_{i=1}^{n} H(z_i, \theta_*)$ is nonsingular w.p.a.1), we write

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \left(n^{-1} \sum_{i=1}^{n} H(z_i, \theta_*) \right)^{-1} \left(-n^{-1/2} \sum_{i=1}^{n} s(z_i, \theta_0) \right)$$

- By the CLT,

$$-n^{-1/2} \sum_{i=1}^{n} s(z_i, \theta_0) \to_d \mathcal{N}(-\mathrm{E}(s(z, \theta_0)), \mathrm{Var}(s(z, \theta_0))).$$

- By the interchangeability of derivative and expectation (integral) and the FOC, we have

$$\mathrm{E}(s(z, \theta_0)) = \nabla_\theta \mathrm{E}(q(z, \theta_0)) = 0.$$

- Similarly, we have A is positive definite,

$$A = \mathrm{E}\left(\frac{\partial^2 q(z,\theta_0)}{\partial\theta\partial\theta'}\right) = \frac{\partial^2 \mathrm{E}(q(z,\theta_0))}{\partial\theta\partial\theta'}.$$

- Letting $B = \mathrm{Var}\, s(z,\theta_0) = \mathrm{E}(s(z,\theta_0)s(z,\theta_0)')$, we conclude

$$n^{-1/2}\sum_{i=1}^{n} s(z_i,\theta_0) \to_d \mathcal{N}(0,B).$$

- By the Slusky lemma, we conclude

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \to_d A^{-1}\left(n^{-1/2}\sum_{i=1}^{n} s(z_i,\theta_0)\right)$$

  hence

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \to_d \mathcal{N}(0, A^{-1}BA^{-1}).$$

  Alternatively, one can show this by $o_p(1)O_p(1) = o_p(1)$.

- The asymptotic variance of $\hat{\theta}_n$ is $n^{-1}A^{-1}BA^{-1}$, which depends on the expected value of Hessian matrix of $q(z,\theta)$, the variance of score $\nabla_\theta q(z,\theta)$ (both evaluated at θ_0), and the sample size. Note in general, $A \neq B$. Though in the MLE case, we have the "information equality" which shows $A = B$.

4 Estimating the asymptotic variance

Components of asymptotic variance

- The asymptotic variance of $\hat{\theta}_n$ is $n^{-1}A^{-1}BA^{-1}$, where

$$A = \mathrm{E}(H(z,\theta_0)) = \mathrm{E}\left(\frac{\partial^2 q(z,\theta_0)}{\partial\theta\partial\theta'}\right),$$

$$B = \mathrm{E}(s(z,\theta_0)s(z,\theta_0)') = \mathrm{E}\left(\frac{\partial q(z,\theta_0)}{\partial\theta}\frac{\partial q(z,\theta_0)}{\partial\theta}'\right).$$

- By lemma 2, we can estimate A and B by

$$\hat{A} = n^{-1}\sum_{i=1}^{n} H(z_i,\hat{\theta}_n),$$

$$\hat{B} = n^{-1}\sum_{i=1}^{n} s(z_i,\hat{\theta}_n)s(z_i,\hat{\theta}_n)'.$$

- Usually, $z_i = (y_i, x_i')'$, e.g. NLS. We can write

$$A = \mathrm{E}(H(z, \theta_0)) = \mathrm{E}(\mathrm{E}(H(z, \theta_0) \mid x)).$$

  Let $A(x, \theta_0) \equiv \mathrm{E}(H(z, \theta_0) \mid x)$. In some cases, $A(x, \theta_0)$ might be easier to estimate. Let $A(x, \hat{\theta}_n)$ be the estimator. Then another estimator of A is

$$\hat{A} = n^{-1} \sum_{i=1}^{n} A(x_i, \hat{\theta}_n).$$

- The asymptotic variance estimator is then

$$n^{-1} \hat{A}^{-1} \hat{B} \hat{A}^{-1} = \left(\sum_{i=1}^{n} H(z_i, \hat{\theta}_n) \right)^{-1} \left(\sum_{i=1}^{n} s(z_i, \hat{\theta}_n) s(z_i, \hat{\theta}_n)' \right) \left(\sum_{i=1}^{n} H(z_i, \hat{\theta}_n) \right)^{-1}$$

Asymptotic variance: NLS example

- For NLS $\mathrm{E}(y \mid x) = m(x, \theta)$, we let $z = (y, x)$ and $q(z, \theta) = (y - m(x, \theta))^2/2$.
- Score:

$$s(z, \theta) = -\nabla_\theta m(x, \theta)(y - m(x, \theta))$$

  Clearly, $\mathrm{E}(s(z, \theta_0)) = 0$.

- Hessian:

$$H(z, \theta) = \nabla_\theta s(z, \theta) = \nabla_\theta m(x, \theta) \nabla_\theta m(x, \theta)' - \nabla_\theta^2 m(x, \theta) u(\theta),$$

  where $u(\theta) = y - m(x, \theta)$.

- Note that $\mathrm{E}(u(\theta_0) \mid x) = 0$. We can estimate $A = \mathrm{E}(H(z, \theta_0))$ by

$$\hat{A} = n^{-1} \sum_{i=1}^{n} \nabla_\theta m(x_i, \hat{\theta}_n) \nabla_\theta m(x_i, \hat{\theta}_n)'$$

- For B, we have

$$\hat{B} = n^{-1} \sum_{i=1}^{n} \hat{u}_i^2 \nabla_\theta m(x_i, \hat{\theta}_n) m(x_i, \hat{\theta}_n)'.$$

- We then have the following heteroskedasticity robust asymptotic variance estimator for the NLS,

$$\left(\sum_{i=1}^{n} \nabla_\theta \hat{m}_i \nabla_\theta \hat{m}_i' \right)^{-1} \left(\sum_{i=1}^{n} \hat{u}_i^2 \nabla_\theta \hat{m}_i \nabla_\theta \hat{m}_i' \right) \left(\sum_{i=1}^{n} \nabla_\theta \hat{m}_i \nabla_\theta \hat{m}_i' \right)^{-1}$$

  Here $\hat{m}_i \equiv m(x_i, \hat{\theta}_n)$.

The bootstrap

- As you have seen the asymptotic variance sometimes can be difficult to compute. The bootstrap is a convenient method for estimating the variance and the distribution of a statistic when the variance formula is difficult to compute.

- Suppose we have iid sample $x_1, \ldots, x_n$ and a test statistic $T_n = g(x_1, \ldots, x_n)$, e.g. an estimator. Let F denote the CDF x.

- Let $v_F(T_n)$ denote the variance of T_n. We have added the subscript F to emphasize that the variance is a function of F. If we knew F, we could at least in principle compute the variance. For example, $T_n = n^{-1} \sum_{i=1}^n x_i$, then

$$v_F(T_n) = \frac{\sigma^2}{n} = \frac{\int x^2 \, \mathrm{d}\, F(x) - (\int x \, \mathrm{d}\, F(x))^2}{n},$$

  which clearly is a function of F.

- If $\hat{F}_n$ is a consistent estimator of F, $v_{\hat{F}_n}(T_n)$ should be a consistent estimator of $v_F(T_n)$.

- Since $v_{\hat{F}_n}(T_n)$ may be difficult to compute, we approximate it with a simulation estimate denoted by $v_{boot}(T_n)$.

- Here is the bootstrap variance estimation procedure:

  1. Draw $x_1^*, \ldots, x_n^* \sim \hat{F}_n$;
  2. Compute $T_n^* = g(x_1^*, \ldots, x_n^*)$;
  3. Repeat steps 1 and 2, B times to get $T_{n,1}^*, \ldots, T_{n,B}^*$;
  4. Let

$$v_{boot} = B^{-1} \sum_{b=1}^{B} (T_{n,b}^* - B^{-1} \sum_{r=1}^{B} T_{n,r}^*)^2.$$

- By the LLN, $v_{boot} \to_{a.s.} v_{\hat{F}_n}(T_n)$ as $B \to \infty$.

- By the convergence of $\hat{F}_n \to F$, we expect $v_{\hat{F}_n}(T_n) \to v_F(T_n)$. (The proof mainly uses the nonparametric delta method.)

- One natural estimator $\hat{F}_n$ is the *empirical cumulative distribution function* (ECDF). It is the CDF that puts mass $1/n$ at each data point x_i,

$$\hat{F}_n(x) = n^{-1} \sum_{i=1}^{n} 1(x_i \leq x).$$

Theorem 3 (Glivenko-Cantelli Theorem)**.** *Let $x_1, \ldots, x_n$ be random draws from F and let $\hat{F}_n$ be the ECDF. Then*

$$\sup_x \left| \hat{F}_n(x) - F(x) \right| \to_{a.s.} 0.$$

Another theorem establishing the nonparametric delta method is useful.

Theorem 4. *If T is Hadamard differentiable with respect to $d(F, G) = \sup_x |F(x) - G(x)|$ then*

$$\sqrt{n}(T(\hat{F}_n) - T(F)) \to_d \mathcal{N}(0, \tau^2),$$

where $\tau^2 = \int L_F(x)^2 \, \mathrm{d}\, F(x)$ and $L_F(x)$ is the influence function of T_n.

- Using the ECDF as $\hat{F}_n$, drawing n points at random from $\hat{F}_n$ is the same as drawing a sample of size n with replacement from the original data. Hence the first step can be replaced by "1. Draw $x_1^*, \ldots, x_n^*$ with replacement from $x_1, \ldots, x_n$."

Median estimation example

- Suppose we have iid draws $u_1, \ldots, u_n$ with n being odd from the uniform distribution $U[0, a]$. Letting $k = (n+1)/2$, it can be shown that

$$\mathrm{Var}(\widehat{median}(u_1, \ldots, u_n)) = \frac{a^2}{4 \cdot (2k+1)}.$$

- Here is an R program illustrating the bootstrap estimation of the variance.

```
set.seed(1011)
N <- 21
k <- (N + 1) / 2
u <- runif(n = N, min = 0, max = 10)
# theoretical sd
sd0 <- sqrt(10^2 / (4 * (2 * k + 1)))
# bootstrap
B <- 50
medianBootstrap <- c()
for (b in 1 : B) {
# draw u with replacement
ustar <- sample(u, size = N, replace = TRUE)
medianBootstrap[b] <- median(ustar)
}
sdbootstrap <- sd(medianBootstrap)
sdbootstrap
# 1.170141
sd0
# 1.042572
```

5 Inference

Given the asymptotic normality and asymptotic variance estimators, it is easy to do statistical inference. Here we study three mostly common test statistics, Wald, LM and LR test statistics.

Single restriction

- To test single restriction like $H_0 : \theta_{0j} = a_j$. We can use

$$t(\hat{\theta}_j, a_j) = \frac{\hat{\theta}_j - a_j}{\text{se}(\hat{\theta}_j)}.$$

  Because $(\hat{\theta}_j - \theta_{0j})/\text{se}(\hat{\theta}_j) \to_d \mathcal{N}(0,1)$, where the standard errors are obtained as the square roots of the diagonal elements of the asymptotic variance matrix, $t(\hat{\theta}_j, a_j) \to_d \mathcal{N}(0,1)$ under the null.

Multiple, nonlinear restrictions: Wald statistic

- We can test multiple, nonlinear restrictions using the asymptotic normality of $\hat{\theta}_n$ and delta method. Consider

$$H_0 : c(\theta_0) = 0,$$

  where $c : \Theta \mapsto \mathbb{R}^q$. Assume c is continuously differentiable on $\text{int}(\Theta)$. Let $C(\theta) = \nabla_\theta c(\theta)$ be the $q \times p = \dim(\theta)$ Jacobian matrix. Define $C \equiv C(\theta_0)$.

- By the mean value theorem argument,

$$\begin{aligned}\sqrt{n}(c(\hat{\theta}_n) - c(\theta_0)) &= \nabla_\theta c(\theta_0)\sqrt{n}(\hat{\theta}_n - \theta_0) + o_p(1) \\ &= C\sqrt{n}(\hat{\theta}_n - \theta_0) + o_p(1).\end{aligned}$$

- Under the null,

$$\sqrt{n}c(\hat{\theta}_n) \to_d \mathcal{N}(0, \Sigma \equiv CA^{-1}BA^{-1}C').$$
$$\sqrt{n}c(\hat{\theta}_n)'\Sigma^{-1}\sqrt{n}c(\hat{\theta}_n) \to_d \chi^2_q.$$

- The Wald statistic is obtained from replacing A, B and C with their consistent estimators,

$$W = nc(\hat{\theta}_n)' \left(\hat{C}\hat{A}^{-1}\hat{B}\hat{A}^{-1}\hat{C}'\right)^{-1} c(\hat{\theta}_n),$$

  where $\hat{C} = C(\hat{\theta}_n)$.

- Under the null, $W \to_d \chi^2_q$.

- The Wald relies on the estimation of the unrestricted model. It is easily available.

Score statistic or Lagrange multiplier (LM) statistic

- Unlike Wald, score statistic is based on restricted estimate. Let $\tilde{\theta}_n$ be the restricted estimate from
$$\min_{\theta\in\Theta}\sum_{i=1}^{n} q(z_i,\theta) \quad s.t. \quad c(\theta)=0.$$

- Under the null, we have the mean value expansion for the score,
$$n^{-1/2}\sum_{i=1}^{n} s(z_i,\tilde{\theta}_n) = n^{-1/2}\sum_{i=1}^{n} s(z_i,\theta_0) + A\sqrt{n}(\tilde{\theta}_n-\theta_0)+o_p(1)$$
Due to the presence of constraints, unlike the unconstrained score $\sum_{i=1}^{n} s_i(z_i,\hat{\theta}_n)=0$, the constrained score $\sum_{i=1}^{n} s_i(z_i,\tilde{\theta}_n)\neq 0$ in general due to the presence of contraints. Also,
$$\sqrt{n}c(\tilde{\theta}_n)=\sqrt{n}c(\theta_0)+C\sqrt{n}(\tilde{\theta}_n-\theta_0)+o_p(1).$$

- Note $c(\theta_0)=0$ by the null and $c(\tilde{\theta}_n)=0$ by the FOC of the Lagrange multiplier. We conclude that $C\sqrt{n}(\tilde{\theta}_n-\theta_0)=o_p(1)$.

- Hence, under the null,
$$CA^{-1}\left(n^{-1/2}\sum_{i=1}^{n} s(z_i,\tilde{\theta}_n)\right) = CA^{-1}\left(n^{-1/2}\sum_{i=1}^{n} s(z_i,\theta_0)\right)+o_p(1).$$

- Recall $n^{-1/2}\sum_{i=1}^{n} s(z_i,\theta_0)\to_d \mathcal{N}(0,B)$. Hence
$$CA^{-1}\left(n^{-1/2}\sum_{i=1}^{n} s(z_i,\tilde{\theta}_n)\right)\to_d \mathcal{N}(0,CA^{-1}BA^{-1}C').$$

- Then the quadratic form
$$LM=\left(\sum_{i=1}^{n}\tilde{s}_i\right)'\tilde{A}^{-1}\tilde{C}'\left(\tilde{C}\tilde{A}^{-1}\tilde{B}\tilde{A}^{-1}\tilde{C}'\right)^{-1}\tilde{C}'\tilde{A}^{-1}\left(\sum_{i=1}^{n}\tilde{s}_i\right)n^{-1}\to_d \chi^2_q.$$
All tilde terms are evaluated at $\tilde{\theta}_n$.

Quasi-likelihood ratio (QLR) statistic

- QLR test statistic is based on the difference of the objective function evaluated at the restricted and unrestricted estimates,

- Let $\tilde{\theta}_n$ and $\hat{\theta}_n$ be the restricted and unrestricted estimates. Using a second-order Taylor expansion, we have

$$\sum_{i=1}^n q(z_i, \tilde{\theta}_n) - \sum_{i=1}^n q(z_i, \hat{\theta}_n) = \left(\sum_{i=1}^n s_i(\hat{\theta}_n)\right)(\tilde{\theta}_n - \hat{\theta}_n) + (1/2)(\tilde{\theta}_n - \hat{\theta}_n)' \left(\sum_{i=1}^n H_i(\theta_*)\right)(\tilde{\theta}_n - \hat{\theta}_n),$$

  where θ_* lies in the line segment connecting $\tilde{\theta}_n$ and $\hat{\theta}_n$.

- By the FOC, $\sum_{i=1}^n s_i(\hat{\theta}_n) = 0$, and we have

$$2\left(\sum_{i=1}^n q(z_i, \tilde{\theta}_n) - \sum_{i=1}^n q(z_i, \hat{\theta}_n)\right) = (\sqrt{n}(\tilde{\theta}_n - \hat{\theta}_n))' A (\sqrt{n}(\tilde{\theta}_n - \hat{\theta}_n)) + o_p(1) \tag{2}$$

- As for $\sqrt{n}(\hat{\theta}_n - \tilde{\theta}_n)$, we use the mean value expansion of $\sum_{i=1}^n s_i(\hat{\theta}_n)$ at $\tilde{\theta}_n$,

$$0 = n^{-1/2} \sum_{i=1}^n s_i(\hat{\theta}_n) = n^{-1/2} \sum_{i=1}^n s_i(\tilde{\theta}_n) + A\sqrt{n}(\hat{\theta}_n - \tilde{\theta}_n) + o_p(1).$$

  Hence

$$\sqrt{n}(\hat{\theta}_n - \tilde{\theta}_n) = A^{-1} n^{-1/2} \sum_{i=1}^n s_i(\tilde{\theta}_n) + o_p(1).$$

  Thus,

$$\text{eq. (2)} = \left(n^{-1/2} \sum_{i=1}^n s_i(\tilde{\theta}_n)\right)' A^{-1} \left(n^{-1/2} \sum_{i=1}^n s_i(\tilde{\theta}_n)\right) + o_p(1). \tag{3}$$

- A **common mistake** is to conclude that

$$n^{-1/2} \sum_{i=1}^n s_i(\tilde{\theta}_n) \to_d n^{-1/2} \sum_{i=1}^n s_i(\theta_0) \to_d \mathcal{N}(0, B).$$

  This is wrong. Under the null $\tilde{\theta}_n \to_p \theta_0$, we can only conclude that

$$n^{-1} \sum_{i=1}^n s_i(\tilde{\theta}_n) \to_p n^{-1} \sum_{i=1}^n s_i(\theta_0) \to_p \mathrm{E}(s_i(\theta_0)) = 0.$$

  However, this does not mean $n^{-1/2} \sum_{i=1}^n s_i(\tilde{\theta}_n)$ and $n^{-1/2} \sum_{i=1}^n s_i(\theta_0)$ have the same asymptotic distribution. Indeed, this is crazy statement, because it means for two consistent estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ for a constant θ_0, they have identical asymptotic

distribution! Here $n^{-1}\sum_{i=1}^n s_i(\tilde{\theta}_n)$ and $n^{-1}\sum_{i=1}^n s_i(\theta_0)$ are two consistent estimators (under the null) of 0. Indeed, we have shown that

$$n^{-1/2}\sum_{i=1}^n s_i(\tilde{\theta}_n) = n^{-1/2}\sum_{i=1}^n s_i(\theta_0) + A\sqrt{n}(\tilde{\theta}_n - \theta_0) + o_p(1).$$

Since $A\sqrt{n}(\tilde{\theta}_n - \theta_0) = O_p(1)$, we can not ignore its role in the asymptotic distribution of $n^{-1/2}\sum_{i=1}^n s_i(\tilde{\theta}_n)$.

- If $A = B$ (*generalized information matrix equality*), eq. (3), hence QLR, has the asymptotic distribution of χ^2_q. (See page 424 of Wooldridge for some matrix algebra arguments.)

6 Two-step estimation

Issues of two-step estimations

- A two-step estimator is one that depends on some preliminary, "first-step" estimator of parameters. The "first-step" can also be two or more step estimation.
- Examples include weighted OLS/NLS, efficient GMM estimation (the first step is about the weighting matrix), generated regressors.
- Question is whether the estimation of the first step affects the asymptotic variance of the second, and if so, what effect does the first step have.
- Ignoring the sampling error in the first step estimation might lead to incorrect confidence interval of the second step estimates.

Set up and consistency of two-step M estimation

- Let $\hat{\gamma}$ be a consistent estimator of a J dimensional parameter γ_*. A two-step M estimator $\hat{\theta}_n$ solves

$$\min_{\theta\in\Theta} n^{-1}\sum_{i=1}^n q(z_i, \theta; \hat{\gamma}).$$

- Under regularity conditions, including continuity of q in θ and γ, the key condition for consistency of $\hat{\theta}_n$ is that θ_0 uniquely solves

$$\min_{\theta\in\Theta} \mathrm{E}(q(z, \theta; \gamma_*)).$$

- This follows from using lemma 2.
- Asymptotic variance is more interesting.

Example: weighted NLS

- Suppose we specify $\mathrm{Var}(y|x) = h(x,\gamma) > 0$. We have a $\hat{\gamma} \to_p \gamma_*$ for some γ_*. For example, let $\tilde{u}_i = y_i - m(x_i, \tilde{\theta})$, where $\tilde{\theta}$ is the NLS estimate. Then $\hat{\gamma}$ solves

$$\min_{\gamma \in \Gamma} n^{-1} \sum_{i=1}^{n} (\tilde{u}_i^2 - h(x_i, \gamma))^2$$

- The weighted NLS (WNLS) estimator solves

$$\min_{\theta \in \Theta} n^{-1} \sum_{i=1}^{n} (y_i - m(x_i, \theta))^2 / h(x_i, \hat{\gamma}).$$

  If $h(x,\gamma)$ is a "good" approximation to $\mathrm{Var}(y|x)$, WNLS might be more efficient than NLS.

- Using lemma 2, we have

$$n^{-1} \sum_{i=1}^{n} (y_i - m(x_i, \theta))^2 / h(x_i, \hat{\gamma}) \to_p \mathrm{E}((y_i - m(x_i, \theta))^2 / h(x_i, \gamma_*)).$$

- If θ_0 uniquely minimizes $\mathrm{E}((y_i - m(x_i, \theta))^2 / h(x_i, \gamma_*))$, $\hat{\theta}_n$ is consistent for θ_0. We can show this.

- Because $\mathrm{E}(y_i - m(x_i, \theta_0) \mid x_i) = 0$, we conclude that for any x_i, θ_0 solves

$$\min_{\theta \in \Theta} \mathrm{E}\big((y_i - m(x_i, \theta))^2 \mid x_i\big)$$

- Because $h(x,\gamma) > 0$ and is a function of x, we have

$$\begin{aligned} \mathrm{E}((y - m(x, \theta_0))^2 / h(x, \gamma) \mid x) &= \mathrm{E}((y - m(x, \theta_0))^2 \mid x) / h(x, \gamma) \\ &\leq \mathrm{E}((y - m(x, \theta))^2 \mid x) / h(x, \gamma). \end{aligned}$$

  Applying the law of iterated expectations, for any θ and any γ,

$$\mathrm{E}((y - m(x, \theta_0))^2 / h(x, \gamma)) \leq \mathrm{E}((y - m(x, \theta))^2 / h(x, \gamma)).$$

- We then just need to assume uniqueness of θ_0.

- WNLS consistently estimate θ_0 for any positive weighting function (including misspecified ones).

Asymptotic variance of two-step M estimation

- Suppose
$$\sqrt{n}(\hat{\gamma} - \gamma_*) = n^{-1/2} \sum_{i=1}^{n} r_i(\gamma_*) + o_p(1),$$
where $\mathrm{E}(r_i(\gamma_*)) = 0$.

- $r_i(\gamma_*)$ is called "influence function". It describes completely the first order asymptotic properties of $\hat{\gamma}$.

- Usually, $r_i(\gamma_*)$ takes the form $Mh(\gamma_*)$ for some matrix M and function h. For example, if the $\hat{\gamma}$ is an M estimator, we know that
$$\sqrt{n}(\hat{\gamma} - \gamma_*) = A^{-1} n^{-1/2} \sum_{i=1}^{n} s_i(\gamma_*) + o_p(1), \quad A = \mathrm{E}(H(\gamma_*)),$$
where s_i is certain score function, and H is certain Hessian matrix. Then $r_i(\gamma_*) = A^{-1} s_i(\gamma_*)$.

- By the arguments of M estimation, we have the expansion,
$$\sqrt{n}(\hat{\theta}_n - \theta_0) = A^{-1} \left(-n^{-1/2} \sum_{i=1}^{n} s(z_i, \theta_0; \hat{\gamma}) \right) + o_p(1),$$
where $s(z_i, \theta; \gamma)$ is the score $\nabla_\theta q(z_i, \theta; \gamma)$ with respect to θ, and A is the expected Hessian matrix
$$A = \mathrm{E}(\nabla_\theta s(z, \theta_0; \gamma_*)).$$

- The score $s(z_i, \theta_0; \hat{\gamma})$ still contains sampling error.

- Letting $s_i(\theta_0, \hat{\gamma}) = s(z_i, \theta_0; \hat{\gamma})$, we have
$$\begin{aligned} n^{-1/2} \sum_{i=1}^{n} s_i(\theta_0, \hat{\gamma}) &= n^{-1/2} \sum_{i=1}^{n} s_i(\theta_0, \gamma_*) + F\sqrt{n}(\hat{\gamma} - \gamma_*) + o_p(1) \\ &= n^{-1/2} \sum_{i=1}^{n} s_i(\theta_0, \gamma_*) + F n^{-1/2} \sum_{i=1}^{n} r_i(\gamma_*) + o_p(1) \\ &= n^{-1/2} \sum_{i=1}^{n} \Big(s_i(\theta_0, \gamma_*) + F r_i(\gamma_*) \Big) + o_p(1) \end{aligned}$$
where
$$F = \mathrm{E}(\nabla_\gamma s_i(\theta_0, \gamma_*)).$$

- Combining the above results, we have

$$\begin{aligned}\sqrt{n}(\hat{\theta}_n - \theta_0) &= A^{-1}\left(-n^{-1/2}\sum_{i=1}^{n}\big(s_i(\theta_0,\gamma_*) + Fr_i(\gamma_*)\big)\right) + o_p(1)\\ &= A^{-1}\left(-n^{-1/2}\sum_{i=1}^{n} g_i(\theta_0,\gamma_*)\right) + o_p(1),\end{aligned}$$

  where

$$g_i(\theta_0,\gamma_*) = s_i(\theta_0,\gamma_*) + Fr_i(\gamma_*).$$

- Note $\mathrm{E}(g_i(\theta_0,\gamma_*)) = 0$. We can apply the CLT to obtain

$$-n^{-1/2}\sum_{i=1}^{n} g_i(\theta_0,\gamma_*) \to \mathcal{N}(0,D), \quad D = \mathrm{E}(g_i(\theta_0,\gamma_*)g_i(\theta_0,\gamma_*)').$$

  We conclude that

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \to_d \mathcal{N}(0, A^{-1}DA^{-1}).$$

Discussion about the asymptotic variance

- The asymptotic variance of the two-step M estimator $\hat{\theta}_n$ is

$$n^{-1}A^{-1}DA^{-1},$$

  where

$$\begin{aligned}D &= \mathrm{E}(g_i(\theta_0,\gamma_*)g_i(\theta_0,\gamma_*)')\\ &= \mathrm{E}(s_i(\theta_0,\gamma_*)s_i(\theta_0,\gamma_*)') + \mathrm{E}(s_i(\theta_0,\gamma_*)r_i(\gamma_*)')F'\\ &\quad + F\,\mathrm{E}(r_i(\gamma_*)s_i(\theta_0,\gamma_*)') + F\,\mathrm{E}(r_i(\gamma_*)r_i(\gamma_*)')F'.\end{aligned}$$

- Let $B = \mathrm{E}(s_i(\theta_0,\gamma_*)s_i(\theta_0,\gamma_*)')$. If we ignore the sampling error in the first step estimation, the variance will be

$$n^{-1}A^{-1}BA^{-1}.$$

- In some cases, $F = \mathrm{E}(\nabla_\gamma s(z_i,\theta_0,\gamma_*))$ is zero. So it is legitimate to ignore the estimation error in $\hat{\gamma}$.

- In some cases, $s_i(\theta_0,\gamma_*)$ and $r_i(\gamma_*)$ are uncorrelated, that is $\mathrm{E}(s_i r_i') = 0$. Then the variance ignoring the first step estimation error will be too small.

- For asymptotic variance of two-step semiparametric estimators, see Newey (1994, The Asymptotic variance of semiparametric estimators, Econometrica).

Estimating the asymptotic variance of the two-step estimator

- Let
$$\hat{F} = n^{-1} \sum_{i=1}^{n} \nabla_\gamma s_i(\hat{\theta}_n, \hat{\gamma}) \to_p F, \quad \hat{g}_i = s_i(\hat{\theta}_n, \hat{\gamma}) + \hat{F} r_i(\hat{\gamma}).$$
- Because $D = \mathrm{E}(g_i(\theta_0, \gamma_*) g_i(\theta_0, \gamma_*)')$, D can be estimated by
$$n^{-1} \sum_{i=1}^{n} g_i(\theta_0, \gamma_*) g_i(\theta_0, \gamma_*)'.$$
- Then
$$\hat{D} = n^{-1} \sum_{i=1}^{n} \hat{g}_i \hat{g}_i' \to_p D.$$
- $\hat{A} = n^{-1} \sum_{i=1}^{n} H_i(\hat{\theta}_n, \hat{\gamma}) \to_p A$, where $H_i(\hat{\theta}_n, \hat{\gamma}) = \nabla_\gamma s(z_i, \hat{\theta}_n; \hat{\gamma})$.
- A valid variance estimator is $n^{-1} \hat{A}^{-1} \hat{D} \hat{A}^{-1}$.

WNLS example

- In the WNLS case, we have the Hessian
$$\nabla_\gamma s_i(\theta, \gamma) = \nabla_\theta m(x_i, \theta)' \nabla_\theta h(x, \gamma) u_i(\theta) / (h(x, \gamma)^2)$$
for any γ.
- Note $\mathrm{E}(u_i(\theta_0) \mid x_i) = 0$. We have $F = \mathrm{E}(\nabla_\gamma s_i(\theta_0, \gamma_*)) = 0$.
- There is no need to adjust the variance of $\hat{\theta}_n$ for the two-step WNLS.
- Thus, we just use $n^{-1} \hat{A}^{-1} \hat{B}^{-1} \hat{A}^{-1}$ as the asymptotic variance estimator,
$$\left(\sum_{i=1}^{n} \nabla_\theta \hat{m}_i' \nabla_\theta \hat{m}_i / \hat{h}_i \right)^{-1} \left(\sum_{i=1}^{n} \hat{u}_i^2 \nabla_\theta \hat{m}_i' \nabla_\theta \hat{m}_i / \hat{h}_i^2 \right) \left(\sum_{i=1}^{n} \nabla_\theta \hat{m}_i' \nabla_\theta \hat{m}_i / \hat{h}_i \right)^{-1}$$
Here $\hat{m}_i = m(x_i, \hat{\theta}_n)$ and $\hat{h}_i = h(x_i, \hat{\gamma})$.